UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | : | COMPUTER SECURITY |
| COURSE CODE | : | CSC662 |
| EXAMINATION | : | OCTOBER 2026 (MOCK) |
| TIME | : | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 5 printed pages*

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## PART A (80 MARKS)

### QUESTION 1

a) State TWO (2) typical locations for placing network-based IDS sensors.

(2 marks)

b) Explain three preventive network security mechanisms that an organization can deploy, describing how each mechanism protects the network.

(6 marks)

### QUESTION 2

a) Explain how a substitution cipher works and what serves as its key.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.

UMRBVVNWVPREQE

(6 marks)

**Vigenere tableau (A = 0)**

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |

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**QUESTION 3**

a) List two formal security models and state the specific concern each one addresses.

(2 marks)

b) State six points about access control structures, covering the access control matrix, access control lists and capability lists, including one advantage and one disadvantage of each structure.

(6 marks)

**QUESTION 4**

a) Explain the goals of access control and the role of a reference monitor.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

**QUESTION 5**

a) State TWO (2) characteristics of Trojan horse malware.

(2 marks)

b) Describe six (6) techniques used by malware to propagate itself or to hide and maintain control of an infected system.

(6 marks)

**QUESTION 6**

a) Explain how a disclosure threat impacts confidentiality and how a disruption threat impacts availability.

(2 marks)

b) Describe the Samy worm as a malware case study, covering how it injected code, how it spread, and its outcome.

(6 marks)

**QUESTION 7**

a) List TWO (2) stages of the biometric authentication process.

(2 marks)

b) Discuss biometric mechanisms and technologies as an alternative to passwords, covering their definition, categories, and the strengths and weaknesses of specific biometric methods.

(6 marks)

**QUESTION 8**

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a) Explain two password security controls used to protect user accounts.

(2 marks)

b) State six facts about Kerberos and single sign-on as authentication mechanisms, covering their motivation, design criteria, login process and limitations.

(6 marks)

**QUESTION 9**

a) Explain how an intrusion detection system detects and responds to attacks.

(2 marks)

b) Explain the distinction between a vulnerability, a threat and an attack, and state three countermeasure activities that together protect assets against threats.

(6 marks)

**QUESTION 10**

a) State TWO (2) facts about the chain of trust in trusted computing.

(2 marks)

b) Refer to your answer in (a). Explain six (6) facts about the NGSCB/Nexus components, covering the Nexus kernel, the Nexus Computing Agent, and how they relate to the TPM and the operating system.

(6 marks)

**PART B (20 MARKS)**

**QUESTION 1**

a) Explain five points about software insecurity and buffer overflows, covering their root causes, how a stack buffer overflow hijacks execution, and one mitigation technique.

(5 marks)

b) Explain five points about the security mechanisms built into Windows, covering the directory service used to organise resources, the registry hives that hold security data, how access rights are encoded, and why the complexity of these mechanisms can itself be a risk.

(5 marks)

**QUESTION 2**

A computing professional at a company is asked to release a system that stores customers' personal information. Before release, the professional must weigh the traditions of belief about right and wrong conduct, the guiding beliefs of the profession, and the formal rules imposed by the government, and must also follow the ACM code of ethics and professional conduct.

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a) Discuss the distinctions between morals, ethics and laws, and explain how the ACM code guides a computing professional's conduct when handling personal information. Support your discussion with examples from the code.

(5 marks)

b) State five professional conduct obligations from the ACM code that the professional in the scenario should follow when releasing the system that stores customers' personal information.

(5 marks)

**END OF QUESTION PAPER**

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